Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One tree-fitting and branch-tracing record.

Explain candidate errors, learned leaves, missingness and evaluation limits.

Keep one habit: W1–W3/W5 → selected cutoff5task → project tree record → separate fitting retry when needed.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Baselines First

> **My remembered idea, example and correction**

Leakage

> **My remembered idea, example and correction**

Inspect Before Modeling

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main six training rows: (1,0,blocked), (1,1,blocked), (1,0,blocked), (0,1,clear), (0,0,clear), (0,1,blocked). Feature 0 is near; feature 1 is tall. Leaf ties choose clear; candidate-score ties keep the first question. Near makes 1 error and stores near 0 → clear, near 1 → blocked. Tall makes 2 errors with both leaves blocked.

  Validation rows (1,1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear) give tree predictions blocked, clear, clear, blocked: 2/4 correct, missing rows 3 and 4. The blocked baseline also scores 2/4 but misses rows 2 and 4. The separate pictured >=2 and >2 rules differ at count 2, with review for missing. The independent cutoff 5 is a separate stored-rule tracing task.

- Reference for this exercise: open this week's student lesson, "Decision Trees", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

> **My reasoning and evidence**

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main six training rows: (1,0,blocked), (1,1,blocked), (1,0,blocked), (0,1,clear), (0,0,clear), (0,1,blocked). Feature 0 is near; feature 1 is tall. Leaf ties choose clear; candidate-score ties keep the first question. Near makes 1 error and stores near 0 → clear, near 1 → blocked. Tall makes 2 errors with both leaves blocked.

  Validation rows (1,1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear) give tree predictions blocked, clear, clear, blocked: 2/4 correct, missing rows 3 and 4. The blocked baseline also scores 2/4 but misses rows 2 and 4. The separate pictured >=2 and >2 rules differ at count 2, with review for missing. The independent cutoff 5 is a separate stored-rule tracing task.

- For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main six training rows: (1,0,blocked), (1,1,blocked), (1,0,blocked), (0,1,clear), (0,0,clear), (0,1,blocked). Feature 0 is near; feature 1 is tall. Leaf ties choose clear; candidate-score ties keep the first question. Near makes 1 error and stores near 0 → clear, near 1 → blocked. Tall makes 2 errors with both leaves blocked.

  Validation rows (1,1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear) give tree predictions blocked, clear, clear, blocked: 2/4 correct, missing rows 3 and 4. The blocked baseline also scores 2/4 but misses rows 2 and 4. The separate pictured >=2 and >2 rules differ at count 2, with review for missing. The independent cutoff 5 is a separate stored-rule tracing task.

- Reference for this exercise: open this week's student lesson, "Decision Trees", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main six training rows: (1,0,blocked), (1,1,blocked), (1,0,blocked), (0,1,clear), (0,0,clear), (0,1,blocked). Feature 0 is near; feature 1 is tall. Leaf ties choose clear; candidate-score ties keep the first question. Near makes 1 error and stores near 0 → clear, near 1 → blocked. Tall makes 2 errors with both leaves blocked.

  Validation rows (1,1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear) give tree predictions blocked, clear, clear, blocked: 2/4 correct, missing rows 3 and 4. The blocked baseline also scores 2/4 but misses rows 2 and 4. The separate pictured >=2 and >2 rules differ at count 2, with review for missing. The independent cutoff 5 is a separate stored-rule tracing task.

- If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This task supplies a stored cutoff5and a missing-count review policy. It assesses branch tracing, equality and the fitting/evaluation distinction; it does not by itself assess fitting a new tree. State which evidence would be needed to claim training selected or improved the rule. Keep the earlier near/tall training and picture's cutoff2separate.

- A stored fitted rule says spots >=5 predicts spotted; otherwise healthy. Trace4,5,6 and a missing count. Policy: missing count requires review. Can test labels be used to choose this cutoff?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Draw the current Plant Classifier tree and annotate feature, **threshold**, **prediction**, and test **evidence** at each **leaf**.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Verify the candidate scores for **Fitting** a decision tree. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Fitting** a decision tree. Proposed explanation: "A near **branch** must predict blocked". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A supplied tree predicts spotted for spot_count>=3, otherwise healthy.
- Trace 2,3,4 and state boundary.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A tree predicts fern when **leaf**_width_cm <= 3 and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 11 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Rules for this fresh tree: try near (feature 0) before tall (feature 1). Each depth-one candidate splits on 0 versus 1 and predicts the training-majority label within each leaf; a tied leaf predicts clear. Select the fewest total training mistakes, retaining the first candidate on a tie. Fit the baseline to the whole training-majority label, with clear on a tie. Validation never changes either fitted rule.

- New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked), (0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

  Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

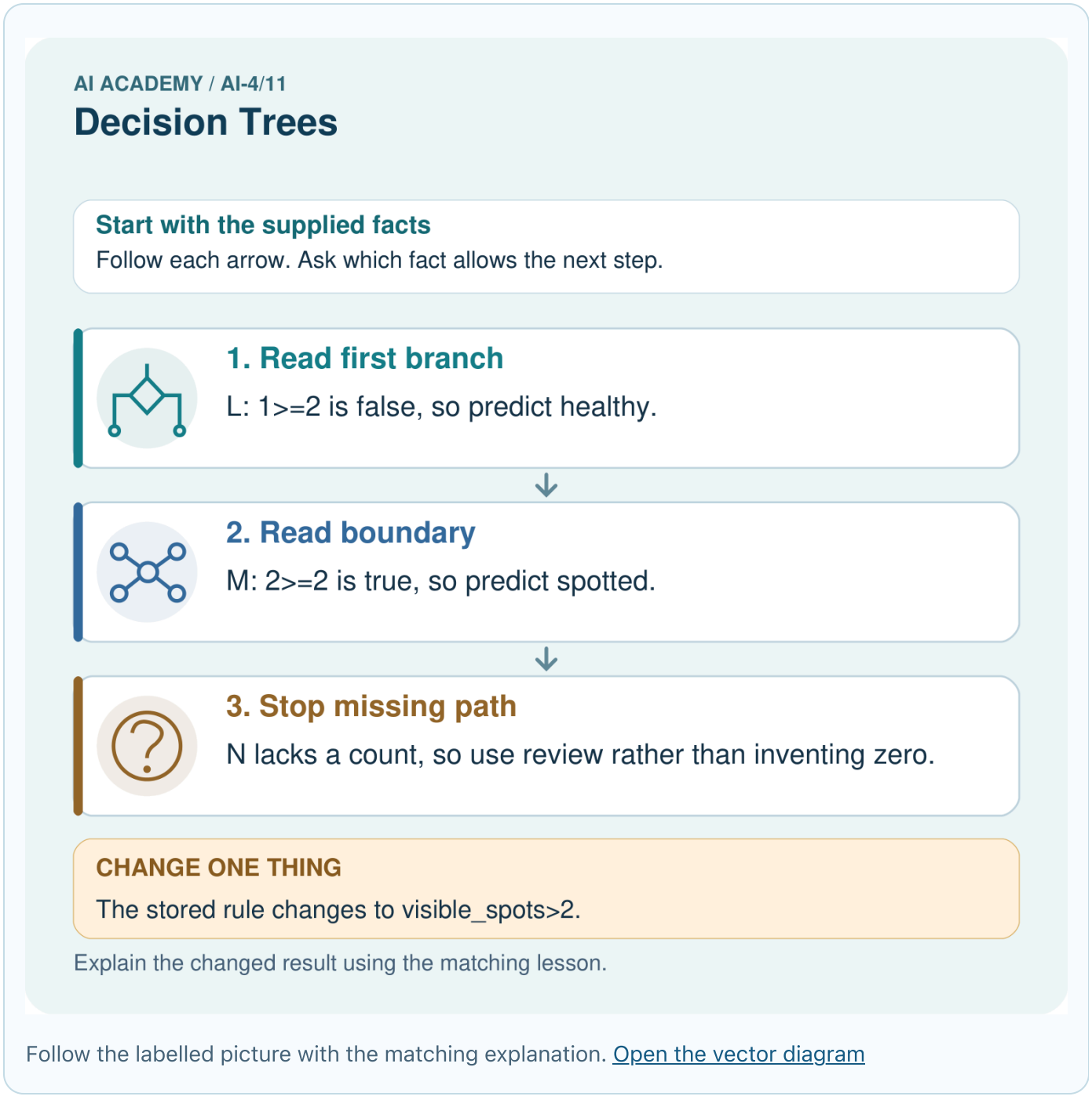
> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Trace a small fitted tree, including its exact boundary.
- Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Trace a small fitted tree, including its exact boundary.
- Which labelled step matches this detail? N lacks a count, so use review rather than inventing zero.
- 1. Read boundary
- 2. Read first **branch**

- 3. Stop missing path

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Stop missing path N lacks a count, so use review rather than inventing zero. Read the picture in order. N lacks a count, so use review rather than inventing zero.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The stored rule changes to visible_spots>2. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. M is now predicted healthy.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: M is now predicted healthy. Changing >= to > moves the exact value 2 to the other **branch**.

Repair and keep the record

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a **split** unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the relevant picture instead of repeating every task.

01. Decision Trees: the case

AI ACADEMY / AI-4/11 · 01 · LOOK AT THE FACTS

Decision Trees: the case

The situation

Trace a small fitted tree, including its exact boundary.

Supplied evidence

Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. Inputs L=1, M=2, N=None.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain decision trees using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

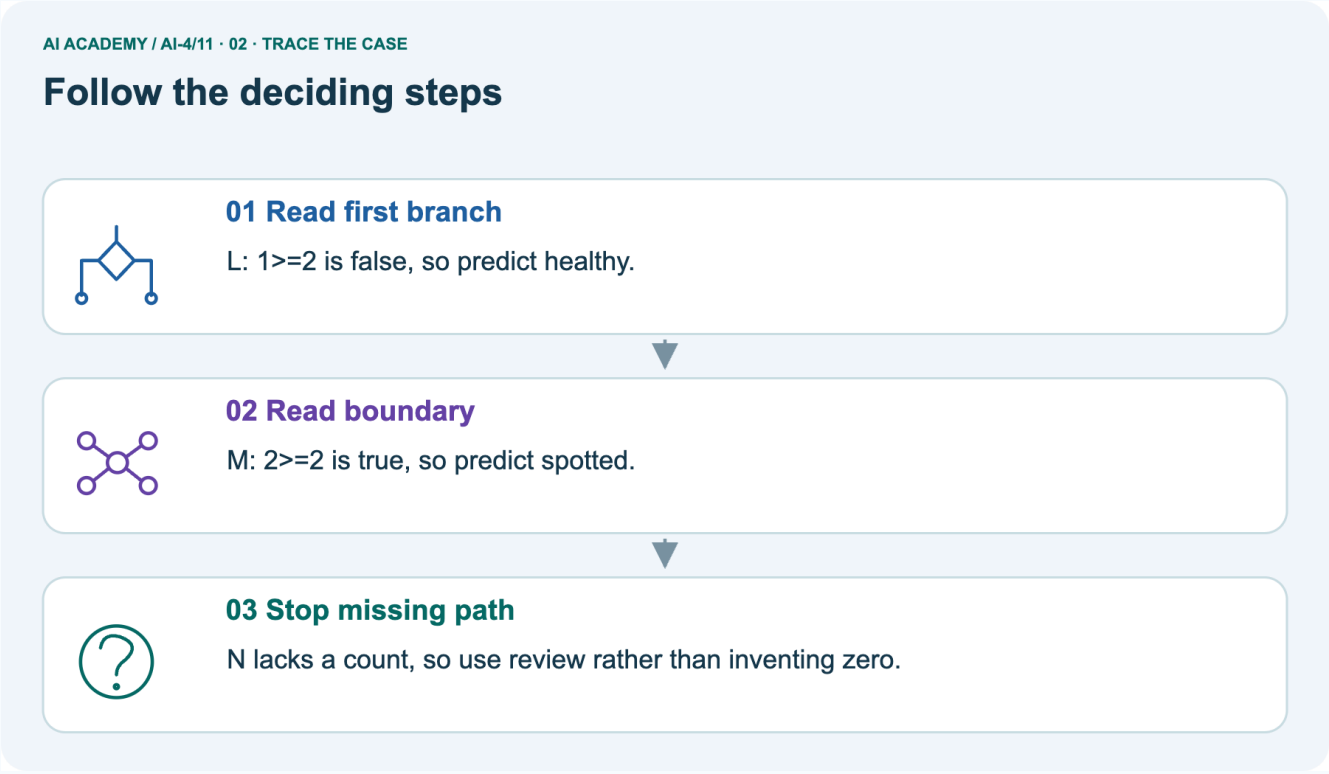
Trace a small fitted tree, including its exact boundary. Supplied facts: Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



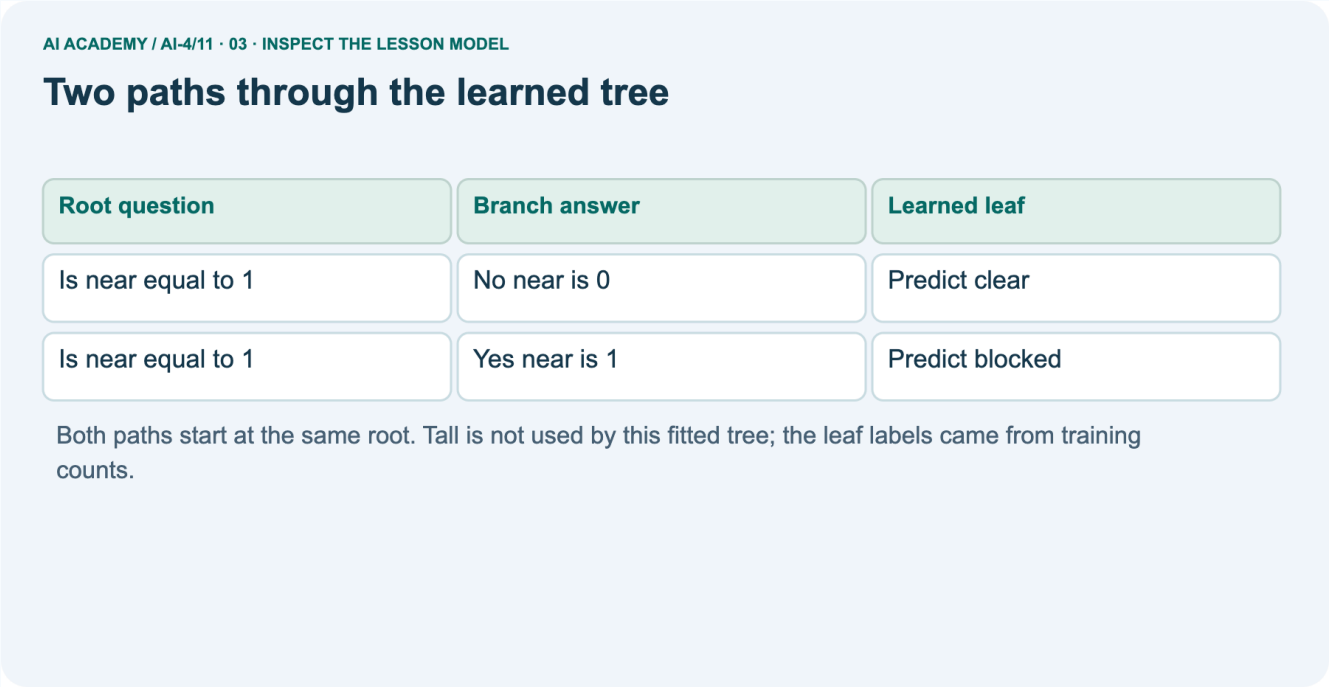
Follow the labelled picture with the matching explanation. Open the vector diagram

Read first **branch**: L: 1>=2 is false, so predict healthy. → Read boundary: M: 2>=2 is true, so predict spotted. → Stop missing path: N lacks a count, so use review rather than inventing zero.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two paths through the learned tree



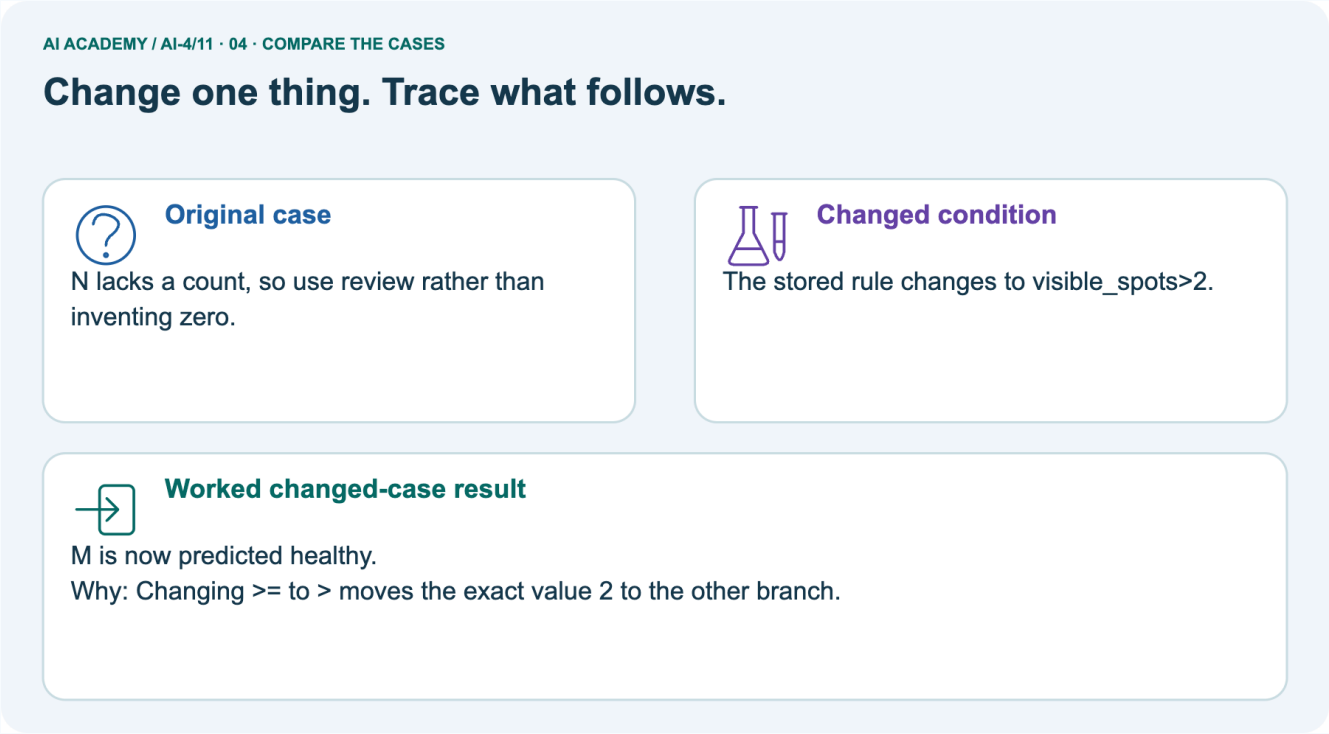
Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

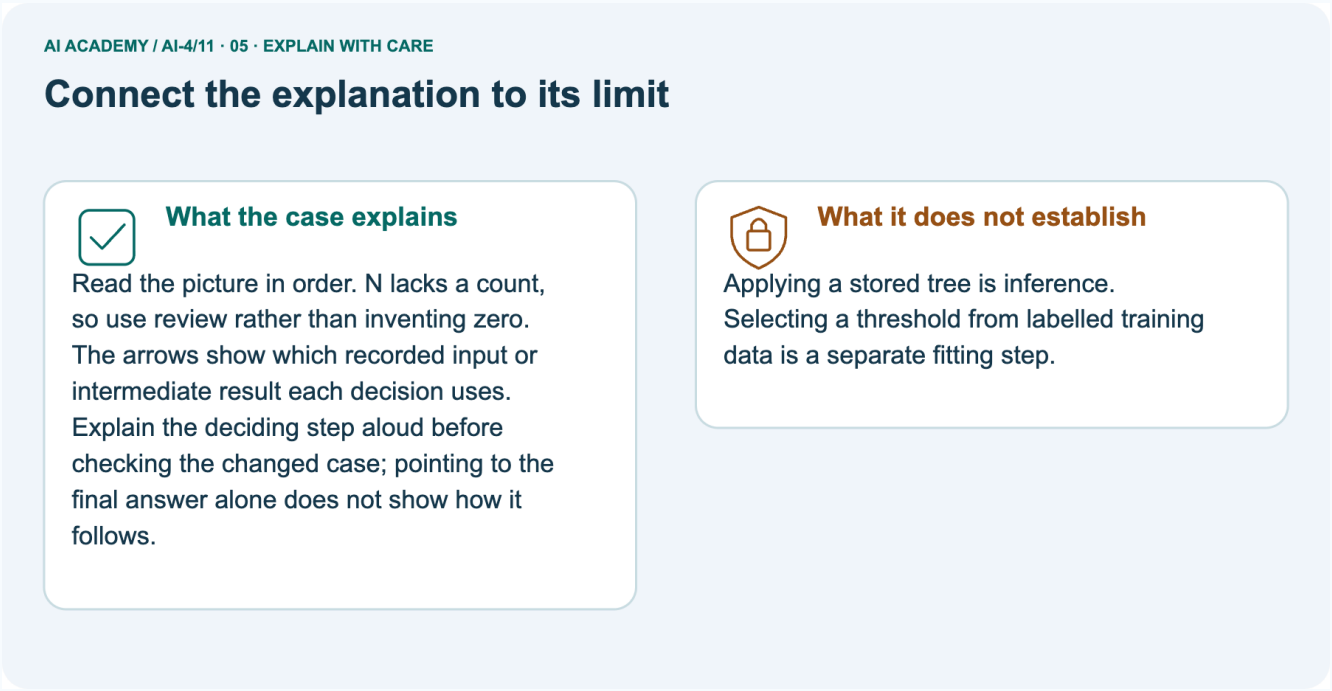
Original case: N lacks a count, so use review rather than inventing zero. Changed condition: The stored rule changes to visible_spots>2. Expected result: M is now predicted healthy.

Reason: Changing >= to > moves the exact value 2 to the other **branch**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

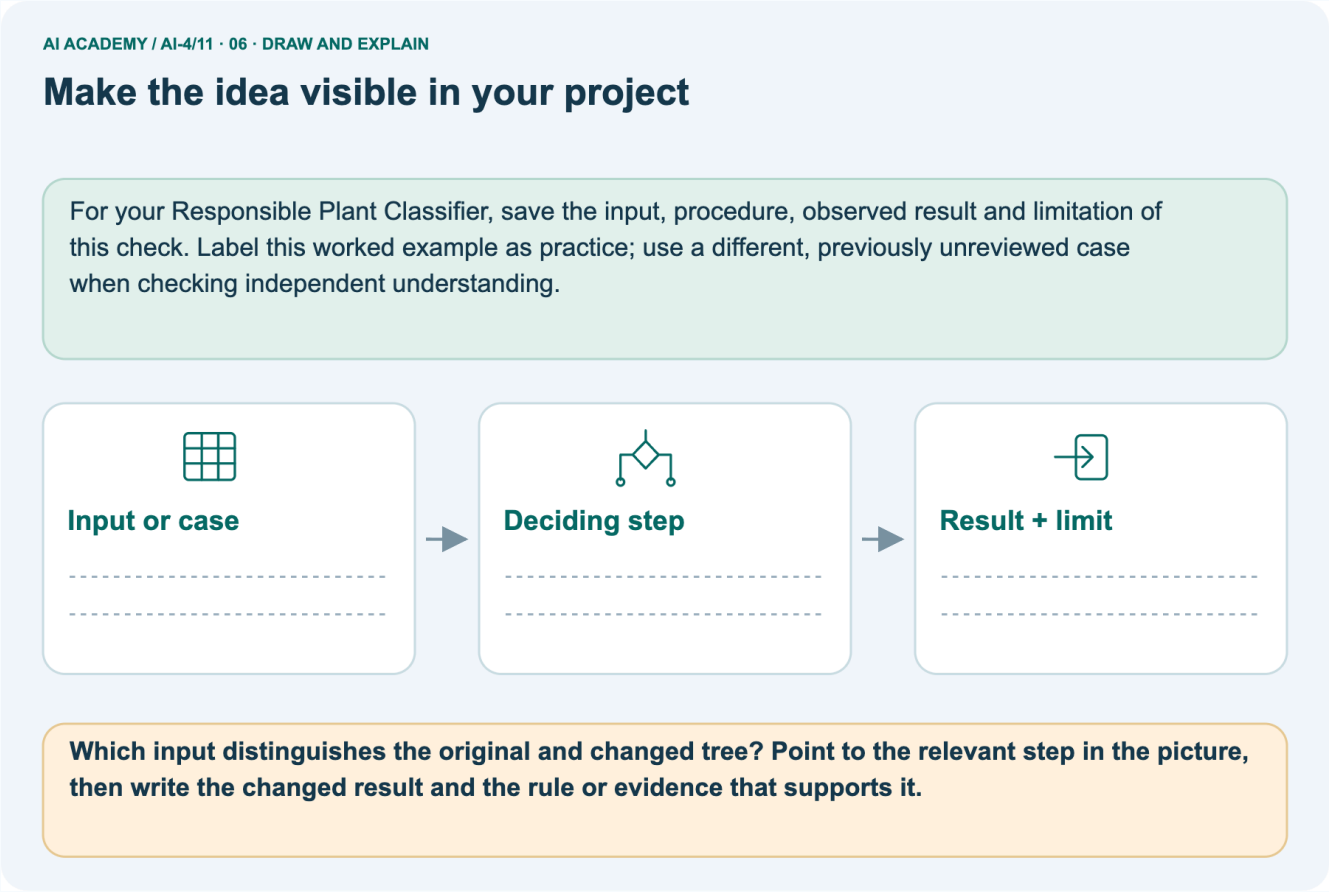
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Applying a stored tree is inference.

Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Decision Trees: the case

02. Follow the deciding steps

03. Two paths through the learned tree

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.